Progress Report

Data Preprocessing Pipeline and Model Direction

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Abstract

This report covers two strands of work on the MIT-BIH arrhythmia data. The first is the data preprocessing pipeline: I reviewed how the literature (six-plus papers) prepares ECG signals, settled on a pipeline of wavelet denoising, per-patient normalization, inter-patient splitting, and fixed window segmentation, and recorded why I preferred a learned dimensionality reduction over PCA and ICA. The second strand is the model direction for the forecasting task, where the chaotic nature of the problem in general, not just the model, limits how far ahead we can predict, and where I argue for rethinking both the model family and the evaluation criterion. Screenshots of the processing applied to the actual data are included as Figures 1 and 2.

1. Literature Review of Preprocessing

I began the data analysis by surveying what the literature already does, gathering six-plus papers whose links are listed in the References. Across them the preprocessing backbone is remarkably consistent. Most pipelines start by removing noise (either with a notch filter, with a combination of high- and low-pass filters, or with a wavelet-based denoiser), then apply normalization, and finally perform window (segment) splitting. A subset of papers add a dimensionality-reduction step, usually Principal Component Analysis (PCA). This common structure, confirmed across both the survey papers [5,6,7] and the applied forecasting and classification work [1,2,3,4], is what I used as the template for my own pipeline.

2. Dimensionality Reduction: Why I Preferred a Learned Approach

PCA. PCA reduces dimensionality by keeping the directions of greatest variance, which means it describes the data purely through its covariance: a second-order, ellipsoidal "shape." That description is complete only for a Gaussian distribution, whose shape is exactly that ellipsoid. ECG is strongly non-Gaussian, with much of its information carried by sparse, high-amplitude deflections (the QRS complex) that are higher-order structure the covariance cannot see. PCA does not strictly require Gaussian data to run, but its notion of "shape" is the Gaussian one, so for a signal this non-Gaussian the leading components are not guaranteed to be the most informative directions. That is why I preferred a learned representation (Section 5) over a linear PCA projection, rather than a claim that PCA cannot work on ECG at all.

ICA. Independent Component Analysis is also common and is effective at isolating and removing noise, but I chose not to build on it either. Its practical limits fit our setting poorly: with only two leads, ICA can recover at most two components, which is weak for separating more than two overlapping sources; its components carry arbitrary sign, scale, and ordering; and it assumes the sources mix linearly, instantaneously, and stationarily. More fundamentally, the independence assumption sits awkwardly on multi-lead ECG, because the leads are largely different projections of the same cardiac dipole and are therefore highly dependent, so the recovered "independent" components need not correspond to meaningful physiological sources. Deciding which components are noise is also subjective, and discarding a component that is not purely noise can remove genuine signal, not only artifacts [8].

3. The Adopted Preprocessing Pipeline

In chronological order, my pipeline is: wavelet denoising, then normalization per patient, then inter-patient splitting, then fixed window segmentation.

Wavelet denoising. The wavelet transform is, in effect, mathematical magic for this problem: a full treatment of the underlying equations is beyond the scope of this report, but the intuition is simple. The transform decomposes the signal across scales. The coherent, high-energy structure of the heartbeat concentrates into a few large coefficients, whereas random noise, being random, spreads thinly across many small coefficients at many frequencies. Thresholding away the small coefficients therefore strips out the noise while keeping the most powerful part of the signal, the true waveform. Figure 1 shows this on real data, including the removal of baseline wander.

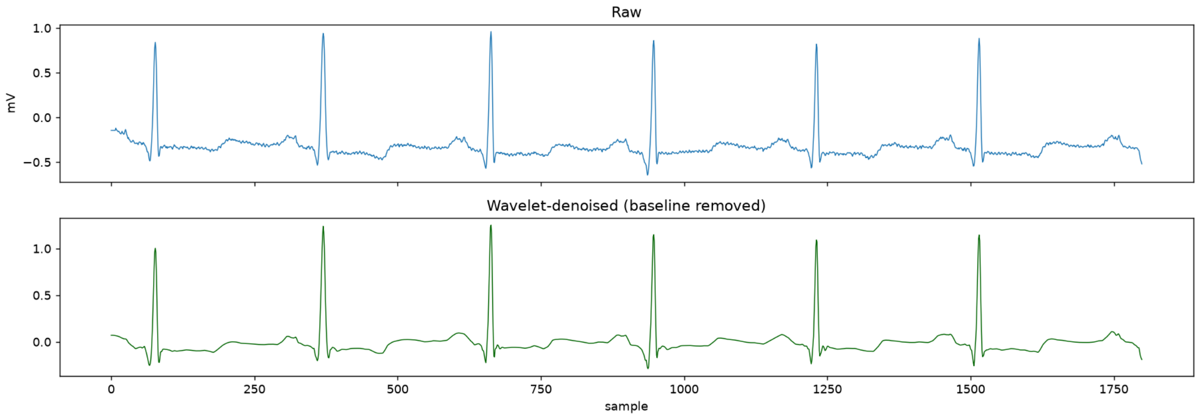

Figure 1: Screenshot of the wavelet denoising applied to the data (record 100, first five seconds): raw signal (top) versus the wavelet-denoised signal with baseline wander removed (bottom).

Normalization (per patient). Normalization is then carried out on every patient alone rather than across the whole dataset. This gives each record a consistent scale (zero mean, unit variance) and, because the statistics never cross between patients, it cannot leak information from one record into another. Figure 2 shows the effect.

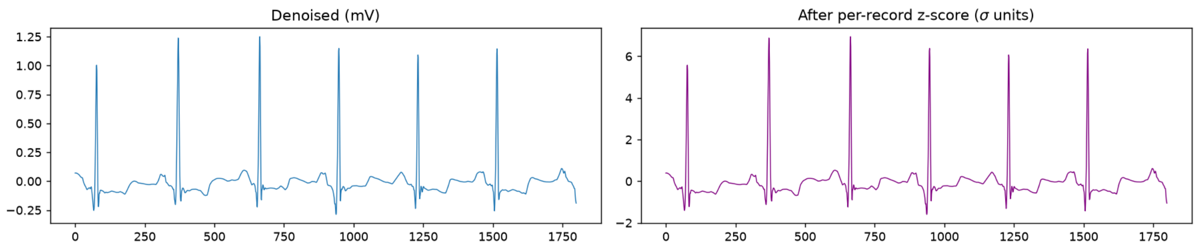

Figure 2: Screenshot of the normalization step (record 100): the denoised signal in millivolts (left) and the same signal after per-patient z-score normalization, now in standard-deviation units (right).

Inter-patient splitting. The normalized records are then split at the patient level (discussed in Section 4).

Window segmentation. Finally, the data is divided by fixed window splitting. Here I followed the literature to the bone, using the standard fixed-size segmentation with an 80/10/10 train/validation/test division.

4. Intra- versus Inter-Patient Evaluation

I believe inter-patient evaluation is both the better and the more honest choice, and a review paper in my references makes the same point [7]. The distinction is about how the data is split. In the intra-patient setting the beats themselves are pooled and split, so a given patient's beats can land in both the training and the test set. The model may then have effectively already seen that patient's pattern during training and simply replicates it at test time, which leaks information and inflates the result. Inter-patient evaluation instead holds out whole patients for testing and trains and validates on different ones. That is the true scenario we care about: a new patient arrives on their own, and we test the model on them.

5. TCN Encoder–Decoder for Dimensionality Reduction

Because PCA and ICA each fit this signal poorly for the reasons above, I adopted a *learned* dimensionality-reduction step instead: a temporal convolutional network (TCN) autoencoder. Its encoder compresses each window (for example from 3600 samples down to roughly 600) into a compact representation, so the downstream model trains faster, and it learns that representation from the data rather than assuming a fixed statistical shape or an independence structure. This follows an established line of work that uses convolutional and TCN autoencoders to compress ECG into low-dimensional features for a downstream model; my contribution is the specific configuration and, in particular, how far the compression can be pushed before too much information is lost, which is still under investigation.

6. Future Work: Data Processing

Remaining directions on the preprocessing side are to explore downsampling, to experiment with the window size to find the right value (or else keep the literature's default), and to explore the TCN encoder further.

7. Model Direction and Open Thoughts

Choosing the model is the hard part, and the deeper difficulty is not specific to ECG but to chaotic systems in general, with ECG being one of the simplest cases. Our problem is forecasting: we must predict the signal's values and, ideally, extend the horizon over which those predictions stay accurate. In a chaotic system a tiny difference between the prediction and the truth grows exponentially, at a rate set by the largest Lyapunov exponent, and throws the two onto separate trajectories. That small difference is unavoidable, because both our measurements and the model itself are approximations. The standard predictability scaling makes the consequence precise: the usable horizon grows only like the *logarithm* of the accuracy, so even a model a thousand times more accurate buys only a little more horizon.[1]

The problem therefore lies partly in the model and partly in the nature of chaotic data, which we cannot change. That leaves two options. Either we find a model that essentially cannot mistake, or, more realistically, we change the evaluation itself. Instead of demanding that every predicted number be exact, we can evaluate whether the prediction follows a *similar trajectory that settles onto the same attractor*: the overall behaviour is right even if the individual numbers differ. In clinical terms, we would judge whether the model correctly predicts, say,

[1] The relevant scaling is $T \sim (1/\lambda)\ln(1/\delta)$ for initial error δ and largest Lyapunov exponent λ; I am happy to share the derivation. My own estimate of λ on the MIT-BIH records was modest but positive across every record, consistent with ECG being a comparatively mild case.

that a patient's heart is accelerating in the same way, not claiming the patient is healthy, but capturing the dynamics.

With that framing, I am proposing to borrow from fields that already forecast chaotic, high-dimensional systems. Weather forecasting is the closest analogue, and there the leading systems use distinct architectures: 3D transformers (for example Pangu-Weather) and graph neural networks (for example GraphCast), among others. My proposal is to investigate whether these, alongside reservoir computing (which has a strong track record on chaotic systems such as the Lorenz attractor), can be adapted to our problem:

- Reservoir computing: reading the primary papers to understand its limitations more precisely.
- 3D transformers and graph neural networks from weather forecasting, treated as candidates to adapt rather than established solutions for ECG.

The next step is to study these approaches in depth and to look for a novel way of my own to approach the problem.

References

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